

Supporting Information

Temperature-controlled Transformation of WO₃ Nanowires into Active Facets-exposed Hexagonal Prisms towards Efficient Visible-light-driven Water Oxidation

Debraj Chandra^{1*}, Tomohiro Katsuki¹, Yuki Tanahashi¹, Takanari Togashi², Yuta Tsubonouchi¹, Norihisa Hoshino¹, Zaki N. Zahran¹ and Masayuki Yagi^{1*}

¹Department of Materials Science and Technology, Faculty of Engineering, Niigata University, 8050 Ikarashi-2, Niigata 9050-2181, Japan.

²Faculty of Science, Yamagata University, 1-4-12 Kojirakawa-machi, Yamagata 990-8560, Japan.

*Correspondence to: yagi@eng.niigata-u.ac.jp and chandra@eng.niigata-u.ac.jp

Contents:

Figure S1. Cross-sectional view of SEM images.

Figure S2. N₂ sorption isotherms and pore size distributions.

Figure S3. XRD fitting patterns.

Table S1. Crystalline lattice analysis.

Figure S4. Resonance Raman spectra.

Table S2. Summary of Raman bands observed.

Table S3. Summary of IPCE data for hitherto-reported WO₃ electrodes.

Figure S5. CVs for electrochemical surface area (ECSA) analysis.

Figure S6. SEM images before and after bulk photoelectrolysis.

Figure S7. XRD patterns before and after bulk photoelectrolysis.

Figure S8. XPS spectra recorded on W 4f and O 1s regions before and after bulk photoelectrolysis.

Figure S9. Product analysis during bulk photoelectrolysis for PEC conversion of benzyl alcohol to benzaldehyde.

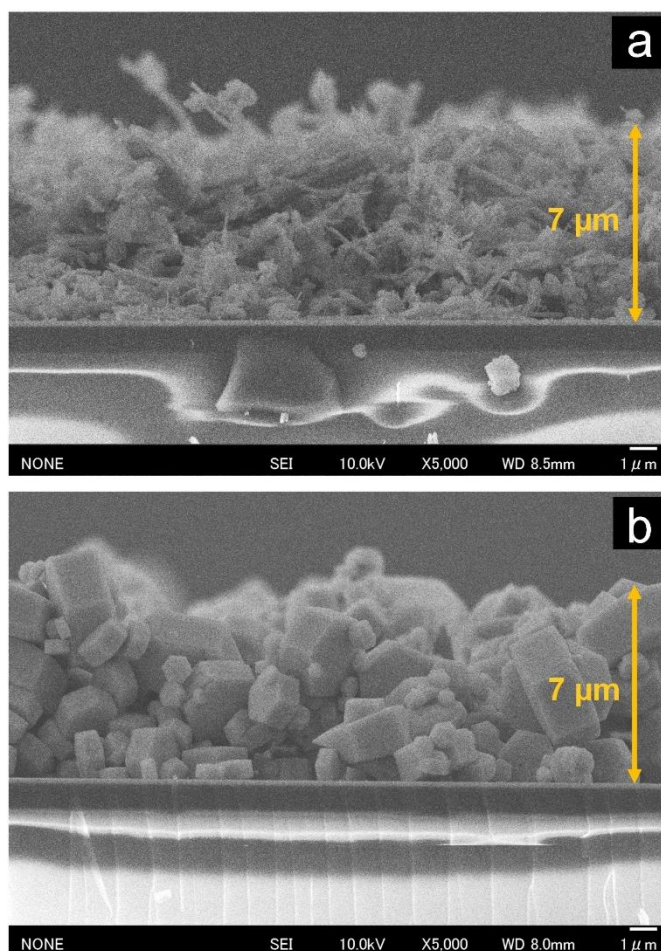


Figure S1. Cross-sectional view of SEM images of (a) NW-WO₃ and (b) HP-WO₃ electrodes.

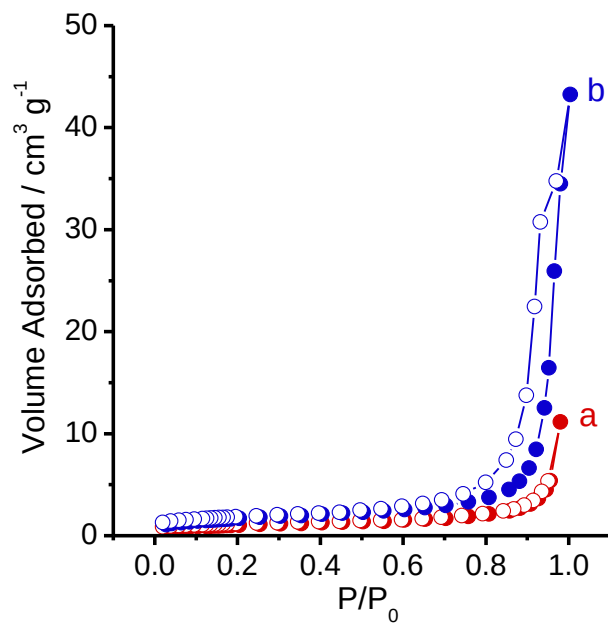


Figure S2. N_2 sorption isotherms of (a) NW- and (b) HP- WO_3 . In N_2 sorption isotherms, adsorption and desorption points are marked by closed and open symbols, respectively.

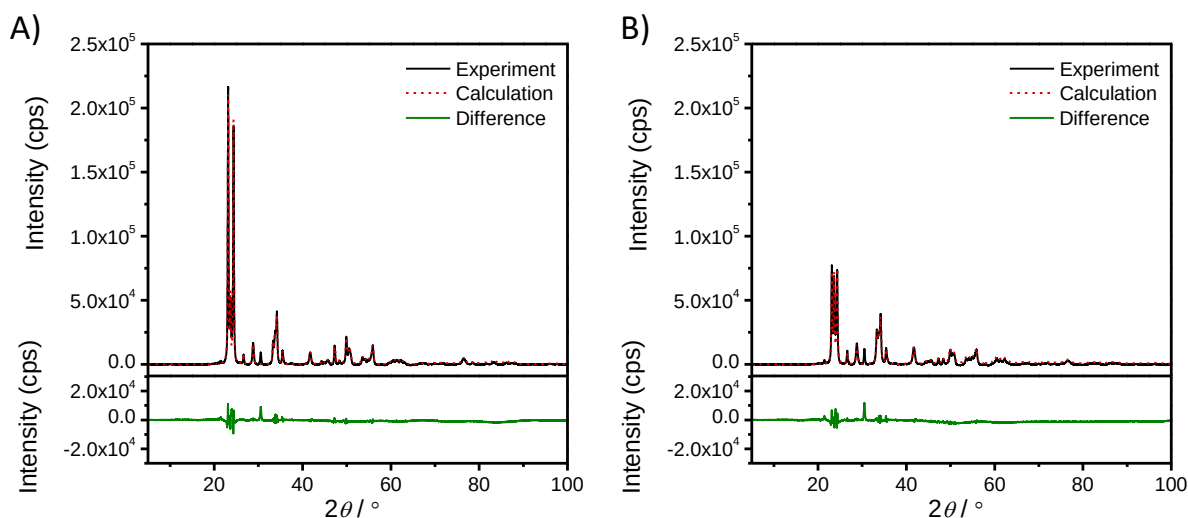


Figure S3. XRD fitting patterns of (A) NW- and (B) HP-WO₃ for experiment (black solid-line), calculation (red dot-line) and difference (green solid-line). The XRD patterns were analyzed by whole-powder-pattern fitting (WPPF) method by PDXL software using a CIF file of monoclinic WO₃ as reference. The S values of NW- and HP-WO₃ for the analysis were 4.0 and 4.3, respectively.

Table S1 Crystalline lattice parameters for NW- and HP-WO₃ calculated from XRD fitting patterns using the WPPF method.

Specimens	a (Å)	b (Å)	c (Å)	α (deg)	β (deg)	γ (deg)	V(Å ³)
Ref. ^a	7.30084(7)	7.53889(7)	7.68962(8)	90	90.892(1)	90	423.19
NW-WO ₃	7.319437	7.536103	7.698277	90	90.45649	90	424.6237
HP-WO ₃	7.3233(4)	7.5284(3)	7.6948(4)	90	90.487(4)	90	424.22(4)

^a analyzed from the JCPDF pattern of monoclinic WO₃ (File number: 01-083-0950)

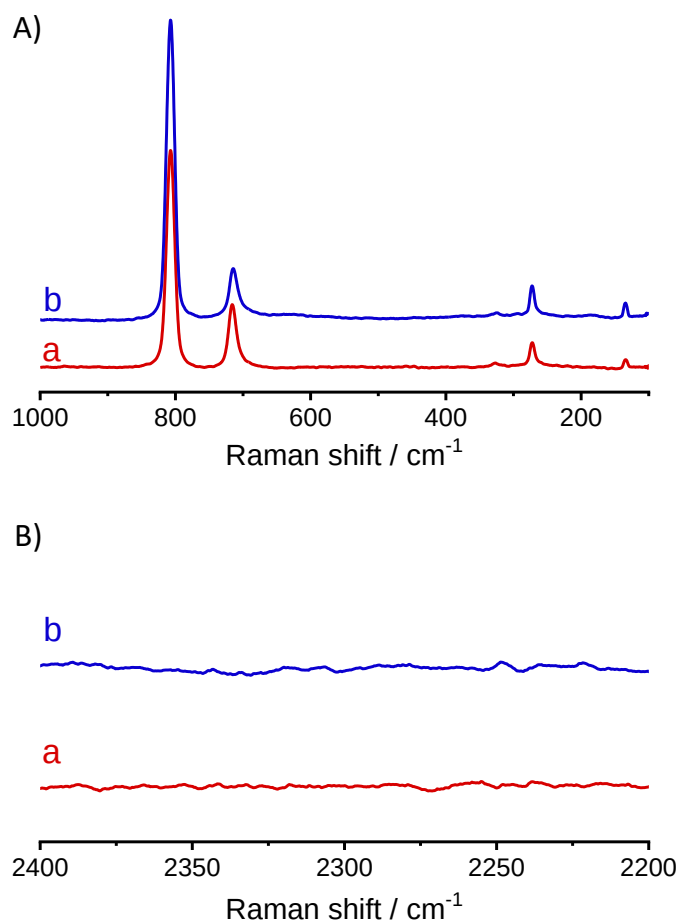


Figure S4. Resonance Raman spectra in the wavenumber region of (A) 1000–100 cm^{-1} and (B) 2400–2200 cm^{-1} for (a) NW- and (b) HP- WO_3 .

Table S2 Summary of Raman shift (cm^{-1}) and assignments for the observed bands for NW- and HP- WO_3 .

Specimens	Raman shift (cm^{-1})			
	Lattice modes	O-W-O bending	W-O stretching	O-W-O stretching
NW- WO_3	133.8	272.2	715.9	807.2
HP- WO_3	134.3	272.6	714.6	807.2

Table S3. Summary of IPCE data (IPCE₄₂₀) at 420 nm reported under the various conditions for hitherto-reported WO₃-based electrodes.

Entry	Structures	IPCE ₄₂₀ (%)	Electrolyte solution	pH	Applied potential	Applied potential (vs RHE)	Reference
1	Hexagonal prism (HP-WO ₃)	47	1 M HClO ₄	0	1.23 V vs RHE	1.23 V	Present Work
2	Nanowire (NW-WO ₃)	15	1 M HClO ₄	0	1.23 V vs RHE	1.23 V	Present Work
3	Mesoporous	32	0.1 M phosphate buffer	6.0	0.5 V vs Ag/AgCl	1.05 V ^{b)}	<i>ACS Sustainable Chem. Eng.</i> , 2018 , 6, 16838.
4	Nanorods N ₂ -intercalation (N ₂ H ₄ -derived) Blocks	43.6	0.1 M phosphate buffer	6.0	0.5 V vs Ag/AgCl	1.05 V ^{b)}	<i>ChemSusChem.</i> , 2018 , 11, 1151.
5	N ₂ -intercalation (NH ₃ -derived) Mesoporous	8.2	0.1 M phosphate buffer	6.0	0.5 V vs Ag/AgCl	1.05 V ^{b)}	<i>ChemSusChem.</i> , 2018 , 11, 1151.
6	N ₂ -intercalation (dodecylamine-derived)	25.4	0.1 M phosphate buffer	6.0	0.5 V vs Ag/AgCl	1.05 V ^{b)}	<i>Chem. Eur. J.</i> , 2017 , 23, 6596.
7	Mesoporous	36	0.1 M phosphate buffer	6.0	0.5 V vs Ag/AgCl	1.05 V ^{b)}	<i>Angew. Chem. Int. Ed.</i> , 2013 , 52, 12606.
8	Nanoplatelets	21	1 M HClO ₄	0 ^{a)}	1.2 V vs SCE	1.44 V ^{b)}	<i>J. Solid State Chem.</i> , 2008 , 181, 175.
9	Nanorods	18	0.5 M Na ₂ SO ₄	7.0	1.23 V vs RHE	1.23V	<i>Small</i> , 2014 , 10, 3692.
10	Nanoflakes	23	0.5 M Na ₂ SO ₄	-	0.8 V vs Ag/AgCl	-	<i>J. Mater. Chem.</i> , 2013 , 1, 3479.
11	Nanoflakes	22	0.1 M Na ₂ SO ₄	-	0.5V vs NHE	-	<i>Nano Lett.</i> , 2010 , 11, 203.
12	Nanoflakes	3	0.1 M Na ₂ SO ₄	7.0	1.2 V vs Ag/AgCl	1.81 V ^{b)}	<i>Chem. Commun.</i> , 2010 , 46, 2769.
13	Nanoparticles	10	0.5 M H ₂ SO ₄	0 ^{a)}	1.2 V vs NHE	1.22 V ^{b)}	<i>Chem. Phys.</i> , 2010 , 123, 225.
14	Nanorods	27	0.5 M Na ₂ SO ₄	-	0.5 V vs NHE	-	<i>Nano Lett.</i> , 2011 , 11, 1928.

15	Nanoparticles	25	1 M H ₂ SO ₄	-0.3 ^{a)}	1.5 V vs. SCE	1.76 V ^{b)}	<i>Langmuir</i> , 2011 , 27, 7276.
16	Nanoplates ((002) facets oriented)	45	0.1 M H ₂ SO ₄	0.7	1.23 V vs RHE	1.23 V	<i>Nano Energy</i> , 2016 , 24, 94.
17	Nanoplates	65	1 M HClO ₄	0 ^{a)}	1.0 V vs RHE	1.0 V	<i>J. Phys. Chem. B</i> , 2001 , 105, 936.
18	Mesoporous	38.5	1 M H ₂ SO ₄	-0.3 ^{a)}	1.0 V vs SCE	1.26 V ^{b)}	<i>Chem. Mater.</i> , 2007 , 19, 5664.
19	Nanoporous	30	0.5 M Na ₂ SO ₄ + 0.1 M HCOONa	-	2.0 V vs Ag/AgCl	-	<i>J. Phys. Chem. B.</i> , 2006 , 110, 25347.
20	Nanoparticles	20	0.5 M NaClO ₄	4.7	1.0 V vs Ag/AgCl	1.48 V ^{b)}	<i>J. Phys. Chem. B.</i> , 2000 , 104, 5686.
21	Nanoporous	18	0.5 M H ₂ SO ₄	0 ^{a)}	1.2 V vs Ag/AgCl	1.40 V ^{b)}	<i>Electrochimica Acta</i> , 2010 , 56, 620.
22	Nanowires	35	0.5 M H ₂ SO ₄	0 ^{a)}	1.2 V vs SCE	1.44 V ^{b)}	<i>Int. J. Hydrogen. Energy</i> , 2009 , 34, 9050.
23	Nanosheets	34	0.1 M Na ₂ SO ₄	-	0.5 V vs Ag/AgCl	-	<i>Chem. Commun.</i> , 2012 , 48, 729.
24	Macroporous	18	0.1 M Na ₂ SO ₄	-	0.8 V vs. SCE	-	<i>ACS Nano</i> , 2011 , 5, 4310.
25	Nanowires	19	0.5 M potassium phosphate buffer	8	1.23 V vs RHE	1.23 V	<i>Nano Lett.</i> , 2014 , 14, 1099.
26	Nanoplates	11	0.2 M Na ₂ SO ₄	7	0.8 V vs Ag/AgCl	1.41 V ^{b)}	<i>Int. J. Hydrogen. Energy</i> , 2016 , 41, 11925.
27	Nanoparticles	28	0.2 M Na ₂ SO ₄	7	0.65 V vs Ag/AgCl	1.26 V ^{b)}	<i>Int. J. Hydrogen. Energy</i> , 2020 , 45, 19257.

^{a)} pH values are calculated from the electrolyte solution conditions.

^{b)} The potentials (E_{RHE}) vs reverse hydrogen electrode (RHE) were calculated by the following equation: $E_{\text{RHE}} = E_{\text{ref}} + (0.059\text{pH}) + E_{\text{ref}}^0$, where E_{ref} is the potentials measured vs Ag/AgCl or SCE, and E_{ref}^0 is the standard potential (0.199 or 0.241 V vs SHE) of an Ag/AgCl or SCE reference electrode, respectively.

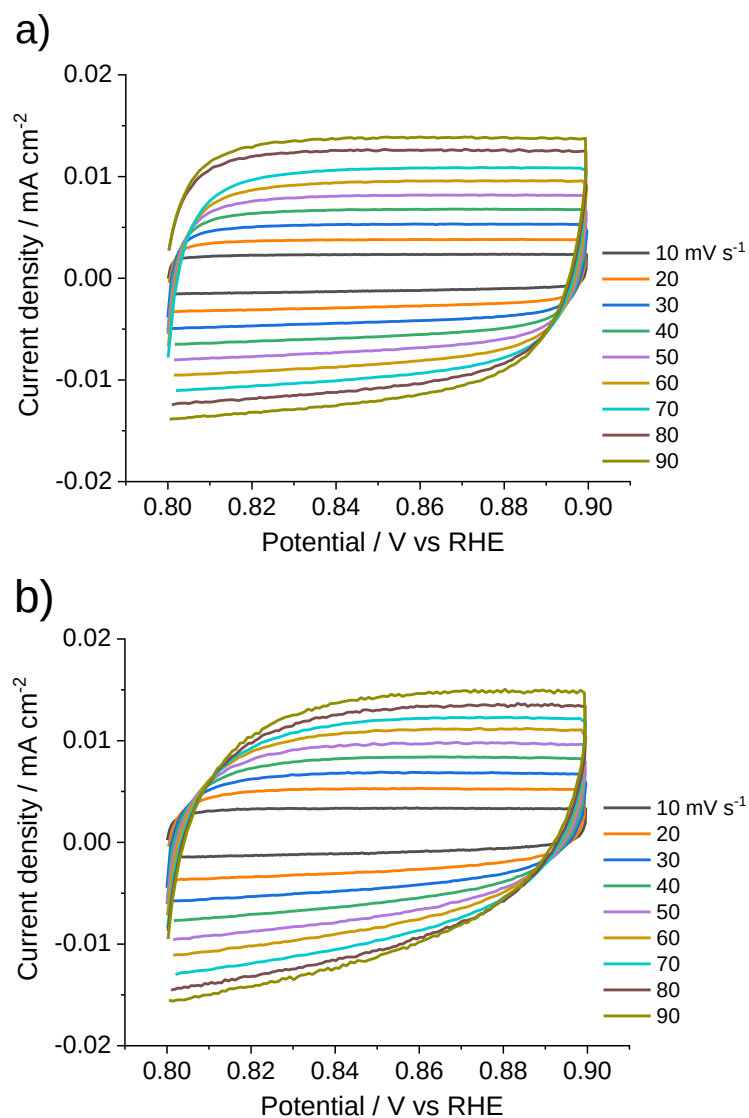


Figure S5. CVs of (a) NW-WO₃ and (b) HP-WO₃ electrodes as measured in 1.0 M HClO₄ solutions (pH = 0) in the potential range of 0.8 – 0.9 V vs. RHE at different scan rates in the range of 10 – 90 mV s⁻¹ under dark conditions.

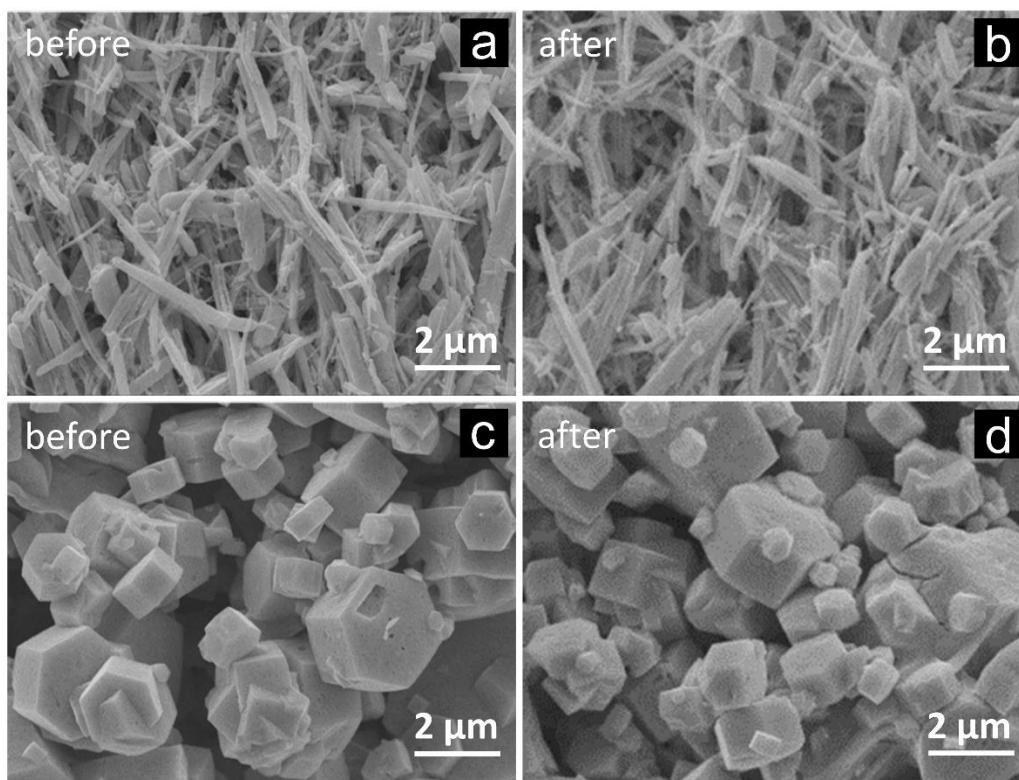


Figure S6. SEM images of (a, b) NW-WO₃ and (c, d) HP-WO₃ electrodes (a, c) before and (b, d) after the 5 h bulk photoelectrolysis.

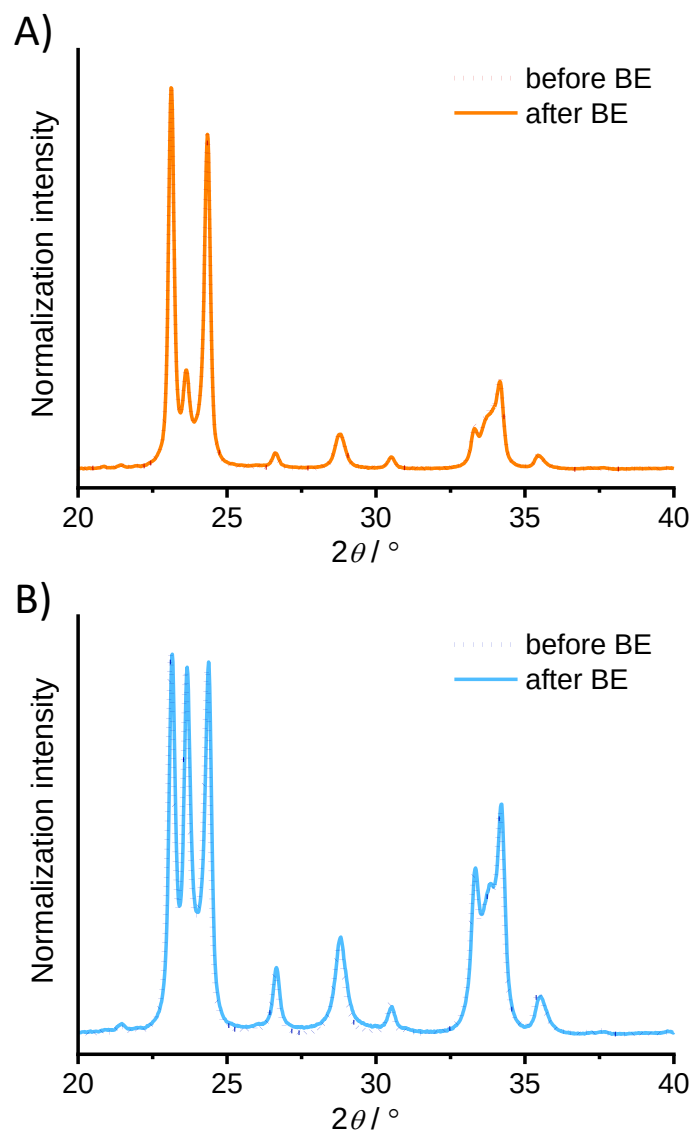


Figure S7. XRD patterns of (A) NW-WO₃ and (B) HP-WO₃ electrodes before (dotted lines) and after (solid lines) the 5 h bulk photoelectrolysis.

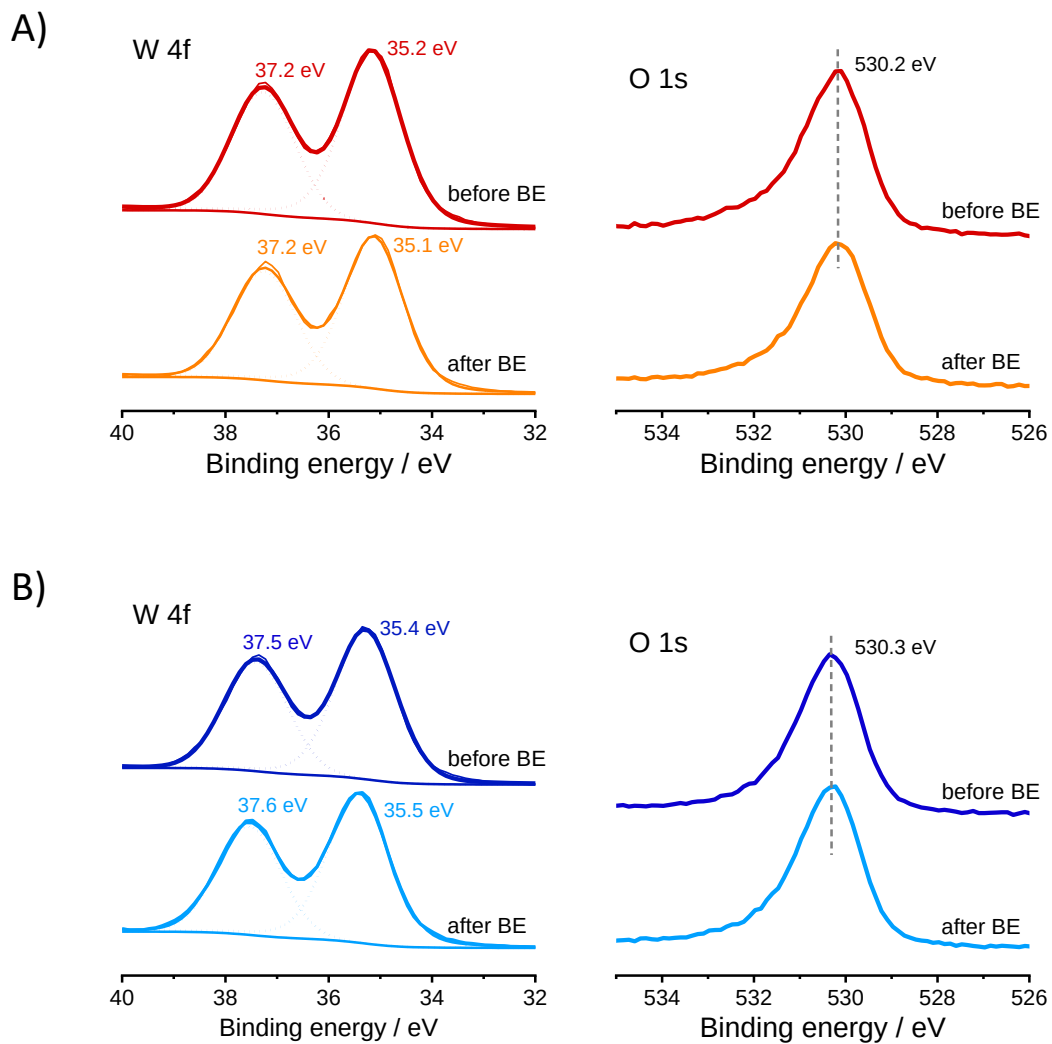


Figure S8. XPS spectra recorded on W 4f and O 1s regions of (A) NW-WO₃ and (B) HP-WO₃ electrodes before (upper) and after (lower) the 5 h bulk photoelectrolysis.

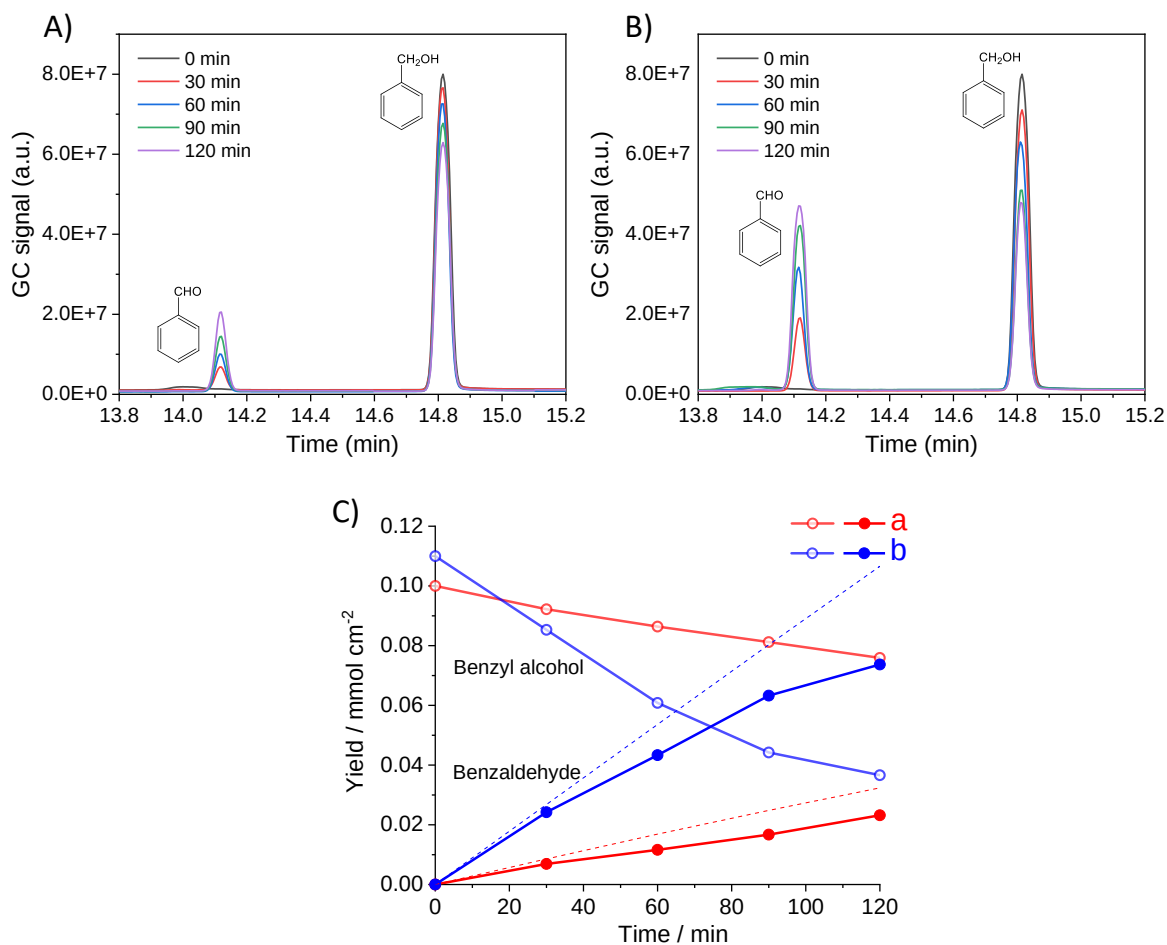


Figure S9. Gas chromatogram changes of the products at different time during bulk photoelectrolysis for PEC conversion of benzyl alcohol to benzaldehyde at 1.23 V vs. RHE using (A) NW-WO₃ and (B) HP-WO₃ electrodes. Peaks at 14.1 and 14.8 of the retention time are assigned to benzaldehyde and benzyl alcohol, respectively. (C) Time courses of amounts of benzaldehyde produced (closed circles) and residual benzyl alcohol (open circles) for the PEC conversion using (a) NW-WO₃ and (b) HP-WO₃ electrodes. Dashed lines are theoretical amounts of benzaldehyde produced, which were calculated from the charges required during the bulk photoelectrolyses.